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We now apply the lemma to the factor group in (4) with

$$Q = G_i \cap H_j, \quad N = G_{i+1}, \quad L = G_i \cap H_{j+1}.$$

Since $Q \cap N = G_{i+1} \cap H_j$, the factor group is isomorphic to

$$(G_i \cap H_j)/(G_i \cap H_{j+1})(G_{i+1} \cap H_j).$$

When we apply the same lemma to the factor group (5), interchanging the roles of the G_k and H_ℓ , we get exactly the same group. Thus the Jordan-Hölder-Schreier theorem follows.

We refer the reader to [1] for the application of the Jordan-Hölder-Schreier theorem to composition series.

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Complex Numbers and the Ham Sandwich Theorem

Andrew Browder

1. INTRODUCTION. The “Ham Sandwich Theorem,” derived from the Borsuk-Ulam theorem, was discussed recently in this MONTHLY [1]. It is the purpose of this note to give a proof of the Borsuk-Ulam theorem (in dimension 2) using only the basic facts about the exponential function on the complex number field $\mathbf{C}$. This article can be considered an addition to my previous MONTHLY article [2].

Throughout this note, we will use the following notation and terminology: $\mathbf{R}$ denotes the real numbers, and $\mathbf{C}$ the complex plane; $\mathbf{C}^*$ signifies the punctured plane $\mathbf{C} \setminus \{0\}$, $D = \{z : |z| \leq 1\}$ the closed unit disk, and $T = \{z : |z| = 1\}$ the boundary of D . We take the two-sphere S^2 to be $\{(x, y, z) \in \mathbf{R}^3 : x^2 + y^2 + z^2 = 1\}$, and for P on S^2 , we denote by P' the point antipodal to P . Thus $(x, y, z)' = (-x, -y, -z)$. If X is any topological space, we let $C(X)$ be the set of all continuous $f : X \rightarrow \mathbf{C}$ and $C^*(X) = \{f \in C(X) : f(X) \subset \mathbf{C}^*\}$, the set of all continuous complex-valued functions on X that never vanish.

If $a \in \mathbf{C}^*$, there exists b in $\mathbf{C}$ such that $e^b = a$; then $e^c = a$ if and only if $c = b + 2n\pi i$ for some integer n . If $f \in C^*(X)$, there need not exist h in $C(X)$ such that $e^h = f$ (for example, if $X = T$ and f is the identity function z). However, we do have the following result:

Proposition 1. *If f belongs to $C^*(D)$, there exists h in $C(D)$ such that $e^h = f$.*

This proposition is our tool in proving the Borsuk-Ulam theorem; in case it is unfamiliar to the reader, we provide a proof in section 3.

2. THE BORSUK-ULAM THEOREM. The theorem we want to prove is:

Theorem 1. *Let $F : S^2 \rightarrow \mathbf{R}^2$ be continuous. Then there exists a pair of antipodal points P and P' such that $F(P) = F(P')$.*

Proof. Identifying $\mathbf{R}^2$ with $\mathbf{C}$, we regard F as an element of $C(S^2)$. Suppose on the contrary that $F(P) \neq F(P')$ for all P in S^2 . Define $G : S^2 \rightarrow \mathbf{C}$ by $G(P) = F(P) - F(P')$. Then G belongs to $C^*(S^2)$, and $G(P) = -G(P')$ for all P in S^2 . We show that no such G can exist, and this contradiction proves the theorem.

Define f in $C(D)$ by $f(x + iy) = G(x, y, \sqrt{1 - x^2 - y^2})$ (i.e., identify the upper hemisphere with the closed unit disk). Then $f(z) \neq 0$ for all z in D —i.e., f belongs to $C^*(D)$ —and $f(z) = -f(-z)$ for all z in T . By Proposition 1, there exists h in $C(D)$ such that $f = e^h$. Let $k(z) = (1/i\pi)[h(z) - h(-z)]$. Then for z in T we have $\exp[i\pi k(z)] = f(z)/f(-z) = -1$; thus $k(z)$ is an odd integer. Since k is continuous, and T is connected, k must be constant. So we have $h(1) = h(-1) + ki\pi = h(1) + 2ki\pi$, forcing $k = 0$, which is impossible since 0 is not an odd integer. ■

3. PROOF OF THE PROPOSITION.

Lemma 1. *Let X be a topological space. If f and g belong to $C(X)$ and $|g| < |f|$, then there exists h in $C(X)$ such that $f + g = fe^h$.*

Proof. Note that the hypothesis implies that f and $f + g$ are never 0. If $r = g/f$, then $|r| < 1$, so $1 + r$ maps X into the right half-plane. It follows that $1 + r = e^h$ for some continuous $h : X \rightarrow \mathbf{C}$. (There exists a continuous branch of $\log z$ in the right half-plane.) Thus $f + g = f(1 + r) = fe^h$, as claimed. ■

Now suppose that f is in $C^*(D)$. Let $\epsilon = \min\{|f(z)| : z \in D\}$. Since f is uniformly continuous on D , there exists $\delta > 0$ such that $|f(z) - f(w)| < \epsilon$ whenever $z, w \in D$ and $|z - w| < \delta$. Choose an integer n such that $n\delta > 1$, and define f_k ($k = 0, \dots, n$) by $f_k(z) = f(kz/n)$ for z in D . Then for $k = 1, \dots, n$ we have

$$|f_k(z) - f_{k-1}(z)| < \epsilon \leq |f_{k-1}(z)|$$

for all z in D . By the lemma, there exist h_k in $C(D)$ such that $f_k = e^{h_k} f_{k-1}$. Now f_0 is the constant function with value $f(0)$, so there is a complex number h_0 with $e^{h_0} = f_0$. Thus we have $f = f_n = e^{h_n} e^{h_{n-1}} \dots e^{h_1} e^{h_0} = e^h$, where $h = \sum_{k=0}^n h_k$. ■

4. COMMENT. Our proof of the Borsuk-Ulam theorem is essentially the proof of Auerbach (as simplified by Steinhaus) given in [1], but it seems simpler in the formulation given here. The referee has observed that “the basic approach is also very similar to those of Chinn-Steinrod [3] and Hatcher [4], though they use the concepts of winding number and fundamental group, respectively.” I would note that winding number involves the notion of homology (working with maps of the circle into a plane set), whereas our approach is a cohomology argument (working with maps from a plane set to the circle), and the fundamental group is of course a notion from homotopy theory. When the plane set is the disk, these three methods are hard to tell apart. By the way, the original proof of Auerbach used the monodromy theorem, a homotopy idea.

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A New Proof of Euclid's Theorem

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A prime number is an integer greater than 1 that is divisible only by 1 and itself. Mathematicians have been studying primes and their properties for over twenty-three centuries. One of the very first results concerning these numbers was presumably proved by Euclid of Alexandria, sometime before 300 B.C. In Book IX of his legendary *Elements* (see [2]) we find Proposition 20, which states:

Proposition. *There are infinitely many prime numbers.*

Euclid's proof (modernized). Assume to the contrary that the set P of all prime numbers is finite, say $P = \{p_1, p_2, \dots, p_k\}$ for a positive integer k . If

$$Q := (p_1 p_2 \cdots p_k) + 1,$$

then $\gcd(Q, p_i) = 1$ for $i = 1, 2, \dots, k$. Therefore Q has to have a prime factor different from all existing primes. That is a contradiction. ■

Today many proofs of Euclid's theorem are known. It may come as a surprise that the following almost trivial argument has not been given before:

New proof. Let n be an arbitrary positive integer greater than 1. Since n and $n + 1$ are consecutive integers, they must be coprime. Hence the number $N_2 = n(n + 1)$ must have at least two different prime factors. Similarly, since the integers $n(n + 1)$ and $n(n + 1) + 1$ are consecutive, and therefore coprime, the number

$$N_3 = n(n + 1)[n(n + 1) + 1]$$

must have at least three different prime factors. This process can be continued indefinitely, so the number of primes must be infinite. ■

Analysis. The proof just given is conceptually even simpler than the original proof due to Euclid, since it does not use Eudoxus's method of "reductio ad absurdum," proof by contradiction. And unlike most other proofs of the theorem, it does not require Proposition 30 of *Elements* (sometimes called "Euclid's Lemma") that states: if p is a prime and $p|ab$, then either $p|a$ or $p|b$. Moreover, our proof is constructive, and it gives integers with an arbitrary number of prime factors.